

# **How degrees of separation affect the preference for indirect harm (and help) in reflective and intuitive judgments.**

Celso Vieira, Department of Philosophy, UFMG (Federal University of Minas Gerais)

## **Author Note**

Celso Vieira <https://orcid.org/0000-0002-3582-2122>

I have no conflict of interest to disclose

Email: [cvb909@gmail.com](mailto:cvb909@gmail.com)

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Ethics Statement Data collection for this study involved a voluntary, fully anonymous online survey assessing altruistic behavior. Participants were informed about the subject, content, and format of the study, and they had to actively consent to participate. They were also informed that they could withdraw at any time. Since the data were collected anonymously and contained no identifying information, the study is in accordance with institutional guidelines for exempt human subjects research.

## **How degrees of separation affect the preference for indirect harm (and help) in reflective and intuitive judgments.**

**Abstract:** People tend to judge indirect harm less harshly than direct ones. While this preference for indirectness is often justifiable, it frequently results in suboptimal decisions as well. For example, a company that ignores the environmental consequences of its actions is judged more harshly than a company that hires another company which ignores environmental consequences even more, even though the latter generates worse results. Interestingly, presenting such conditions together is enough to cancel out the preference for indirect actions. (Paharia et al. 2009). While the preference for indirectness is well documented, most studies neglect an important aspect of how this preference affects the world today. Many of our actions are part of a long causal chain involving some form of harm. A paradigmatic example is the complex and long supply chains of the products we routinely buy (Eriksson et al. 2013). In these so-called “new harms” (Lichtenberg 2010), indirectness occurs in degrees. Consider the different responsibilities of distributors, retailers, and consumers when purchasing inexpensive products that exploit workers. Thus, in addition to the direct/indirect dichotomy, it is important to examine whether and how different degrees of separation affect our moral judgments. Another related yet understudied aspect is whether moral praise and blame behave differently in this context. To start closing these gaps, the present study analyzes moral scores assigned to scenarios depicting actions ranging from 0 to 4 degrees of separation from harm or help. There were two modes of presentation: joint and separate evaluations. The results indicate that degrees matter, particularly in the separate evaluation of harmful causal chains. In contrast, a joint evaluation distributes responsibility more evenly across the chain. Interestingly, there are no degrees of indirectness within the help valence, and not even a difference between direct and indirect actions in the joint evaluation.

## **Introduction**

Imagine finding out that your neighbor keeps their pet in a crate where it does not have enough room to turn around. Now, imagine that you find out your neighbor is boarding their pet in a budget hotel where the animals are kept in crates too small for them to turn around in. Finally, imagine that you find out your neighbor buys eggs from a supermarket that buys them from a factory farm that keeps tens of thousands of animals in crates where they cannot turn around. Chances are, you would judge your neighbor quite differently in these situations.

Real-world situations like the ones described above are messy and contain many factors that contribute to variations in moral judgments. Despite this complexity, the variation between direct and indirect harm is a clear difference that exerts a significant impact on our moral judgments. According to the preference for indirectness, moral judgments of indirect actions tend to be much milder than those of direct actions (Royzman & Baron, 2002; Waldmann & Dieterich, 2007; Cushman, et al., 2006; Greene et al. 2001). While such mitigation may sometimes be justified, it is often not, to the point that many consider it a bias (Royzman & Baron, 2002).

Following the preference for indirectness, people judge a company that increases the price of a medication, thereby causing harm to ill people, much more harshly than a company that strategically makes a deal with another company to distribute the drug at an even higher price, in spite of the fact that the second case is as intentional as the first one and generates more harm. However, when the scenarios are presented together the preference reverses to the most plausible

one, in which the option causing more harm is judged to be worse, despite being indirect (Hsee et al., 1999, Paharia et al. 2009).

The fact that indirectness may unjustifiably mitigate moral judgments is particularly relevant in today's world. After all, as the supermarket scenario above illustrates, many of our daily actions may indirectly contribute to harm further down the line. Supply chains taking advantage of exploitative working conditions far away from the final end consumer provide the best example. This phenomenon is pervasive enough to deserve a label. Lichtenberg (2010) calls them the "new harms". Additionally, since these new harms depend on modern world structures, chances are that our moral judgments – which were shaped in a very different context – are not well prepared to deal with them. In fact, the reverse is probably true. Some of their features may well have been created to exploit our unsuitable heuristics.

It is very likely that our preference for indirectness explains part of our inability to properly address the new harms. However, even if this is the case, a binary contrast of direct versus indirect, which is usually used in research, will fall short of capturing the full picture. After all, as the supply chains exemplify, indirectness comes in degrees in the new harms. Different agents throughout the entire chain, such as distributors, producers, and consumers, may benefit from the harms occurring down the line. In light of our preference for indirectness, it is reasonable to assume that different distances from the harm will affect our moral judgments. Given that the preference for indirect harm leads to suboptimal judgments, a better understanding of the effects of varying degrees of indirectness is instrumental for living a moral life in the current state of the world.

The present study aims to contribute to a more nuanced understanding of the preference for indirectness by gathering evidence about how different degrees of separation influence our moral evaluations. To this end, we present scenarios describing agents from 0 to 4 degrees of separation to a harm. Participants then judge each of these agents behavior using a scale from 0 (morally neutral) to 100 (extremely wrong). Additionally, to examine potential judgment biases, the separate and joint modes of evaluation will be compared. Finally, to counter the tendency to only research harmful behavior in a moral context, positive scenarios involving indirect help will be analyzed.

## **Indirectness**

The preference for indirectness is strongest when directness corresponds to physical contact and its absence corresponds to indirectness. (Royzman & Baron, 2002; Waldmann & Dieterich, 2007; Cushman, et al., 2006). In this context of physical causation, direct action can be conceptualized as an action that causes harm in a way that makes it difficult to separate the action from the harm without redefining the action itself (Royzman & Baron, 2002). For example, contrast a fall caused by pushing someone with a fall caused by removing their support. According to the previous definition, only pushing is a direct cause. However, the perception may change when the focus is on responsibility. Would you not say that the agent who removed the support was directly responsible for the fall?

In normative contexts, people seem to treat causality and responsibility interchangeably (Sytsma 2021, Sytsma & Livengood 2021, Sytsma 2022). Being directly responsible for a harmful outcome is sufficient to be considered the cause, even if one is not physically involved. This conflated attitude toward responsibility and causation suits modern societies, where ways to cause harm without direct involvement are abundant (Paharia et al. 2009). A CEO who makes a decision is perceived as directly responsible for its consequences, even if they do not carry out the action

themselves. In this context, it is not the absence of contact but transferring the decision-making responsibility to someone else that makes the action no longer direct. This responsibility-centered framework suits the new harms, with the addition that the transference of responsibility may involve many agents and degrees of separation. Accordingly, for the purposes of this study, an indirect action is characterized by the presence of a mediator. This allows us to measure the effect of each added degree of indirectness on moral evaluations by introducing more and more mediators into a causal/responsibility chain.

Some studies have found that adding even one mediating agent can have a significant impact on moral judgment and behavior. For example, adding one mediator increased the likelihood that people would obey an order to harm strangers (Milgram 1974). Additionally, introducing a mediating enterprise that increases the price of medicine eased negative evaluations of the producing enterprise (Paharia et al. 2009). In economic games, eliminating a direct link between action and harmful consequence increases self-interested behavior. Even more astonishingly, participants consciously choose to not reveal this link so that they can act in their own interest without feeling guilty (Dana et al. 2003). These behavioral tendencies highlight the importance of understanding how mediation affects moral behavior. However, distinguishing only between direct and mediated is insufficient in the context of new harms. We must also understand how adding a chain of mediators and different degrees of separation affects our moral judgments.

## **New Harms and Moral Disengagement**

The term 'new harms' refers to the myriad ways in which the ordinary activities of modern citizens may cause harm to others due to factors ranging from natural phenomena, such as climate change, to conventional structures, such as the global market (Lichtenberg, 2010). The latter includes even the simplest items we consume, such as chocolate, which tend to involve complex supply chains containing injustices that we would not tolerate if they were happening closer to home. Properly reacting to these harms is particularly challenging because we tend to experience causal and responsibility influence as inversely proportional to spatial and temporal distance (Scheffer 1995). This scenario facilitates moral disengagement, defined as a mechanism used to obfuscate morality when participating in ethically questionable activities (Bandura, 1999; Bandura et al., 1996; Vollum and Buffington-Vollum, 2007).

Supply chains are the paradigmatic example of new harms and moral disengagement in the present study (Burgess et al. 2006, Eriksson et al, 2013a, Eriksson et al, 2013b, Eriksson & Svensson 2016). Although no one wants to harm workers, doing so satisfies various interests along the chain: stakeholders want to increase profits, production managers want to reduce costs, sourcing managers want the cheapest deals with extraction sites, and consumers want accessible goods. This tension creates a prudential interest in overlooking the link between the shared harm and derived advantage (Dana et al. 2003). Furthermore, the multitude of agents involved in the chain at various levels and in different roles multiplies the opportunities to diffuse responsibility by minimizing one's own involvement (Bandura, 1999; Bandura et al., 1996). Hence, it is important to look for mechanisms that combat this moral disengagement.

Moral responsibility in complex scenarios, such as those involving new harms, cannot be captured by a yes/no dichotomy. This is why supply chain investigations treat the chain as a continuous spectrum through which not only money and goods, but also responsibility, should flow (Ashby et al. 2012). Nevertheless, even in this context, a binary approach still prevails. There have been interesting efforts to find the moral decoupling point, which is the point at which moral

responsibility is interrupted, reduced, or distorted (Eriksson & Svensson 2016). This aim is plausible and will be addressed in the present study (see "Accountability" below). Nevertheless, since degrees of separation are one of the defining features of new harms, our understanding will be incomplete without investigating how moral judgments progress throughout the chain and how to avoid pitfalls during this progression.

## **Reflection**

Previous research has shown that preferences for indirectness are susceptible to preference reversal, which is the phenomenon in which humans change their preferences when evaluating options separately or together (Hsee et al., 1999). Consider the following scenarios (after Paharia et al. 2009):

- (i) a well-known real estate company X decided to only clean up 40% of the toxic waste on the land of a new housing development.
- (ii) a well-known real estate company X decided to sell land contaminated with toxic waste to another lesser-known company, Y, that did not engage in any clean up effort.

When presented separately, people are harsher in condemning the direct harm caused by X in (i). However, presenting them together reverses the results. Despite being indirect, X's action in (ii) is judged worse due to its more harmful consequences. This reversal violates the presumption that our choices should be transitive. If (ii) is worse than (i), then it should always be worse, regardless of the mode of presentation.

The proposed explanation for the transitivity failure is that separate evaluations elicit intuitive and affective judgments, whereas the joint mode invites more reflective comparisons (Hsee et al., 2005; Hsee et al., 1999; Bazerman et al., 1998). Since direct harm arouses more emotions, the separate scenario results in harsher condemnation in (i). In contrast, adopting a more reflective attitude allows one to weigh the greater harm caused in (ii).

Whether moral behavior relies on affection or reflection depends on situational factors and the moral principles at play (Cushman et al., 2006). In the context of supply chains, there is a willingness to avoid an emotional public response. Most enterprises accept taking responsibility for how their immediate suppliers treat their workers. However, in order to increase their competitiveness, they are ready to overlook harms occurring further down the supply chain, thereby taking advantage of the disengaging effects of indirectness. In contrast to this standard practice, a holistic view of the supply chain prescribes an ideal situation in which enterprises would take responsibility for the entire chain (Ashby et al. 2012). According to the preference reversal observed when direct and indirect actions are presented together, this holistic view would encourage a more reflective attitude, which could lead to optimal moral behavior.

In short, similar to using joint evaluations to reverse the preference for indirectness, economists suggest that increasing awareness of the entire supply chain would reduce the risk of moral disengagement and encourage responsibility. This is why the present study includes a joint evaluation condition. Perhaps seeing all the degrees in the harm chain will elicit a reflective attitude and mitigate the preference for indirectness, leading to a fairer distribution of responsibility across the whole chain.

## **Valence asymmetry**

The aforementioned studies illustrate that most research on the preference for indirectness focuses on harms. This tendency reflects the typical emphasis on negative-valenced actions characteristic of moral investigations. In this context, negative valence refers to everything that deserves moral blame, with an emphasis on what people should not do, such as harming others. In contrast, a positive valence refers to actions that are morally laudable, such as helping.

The emphasis on the negative is justified because a prescription to refrain from immoral behavior seems more binding than moral recommendations for positive actions, which tend to be optional (Janoff-Bulman et al., 2009, Pavey et al. 2021). For example, helping is generally considered morally good, yet we often act as if it is up to each person to decide whether and when to help. In contrast, individuals do not get to decide when they should refrain from causing harm. This is a moral prohibition. Accordingly, there are many more laws that punish immoral behavior than incentives for good behavior (Feldman, 2018).

Despite the focus on the negative, it is reasonable to question why prohibitions are more binding. One popular explanation relies on the perception that avoiding wrongdoing is less demanding than doing good, as compliance with the latter may violate our autonomy (Woollard, F., 2015). This perception is evident in the case of old harms. While it is fairly easy to refrain from physically attacking people, helping those in need can quickly become a burdensome task. However, we should not jump to conclusions so quickly since the context of new harms is also peculiar in this regard.

We live in a society where most citizens can easily avoid causing physical harm to others. Nevertheless, since the new harms have become so entrenched in our daily routines, it is almost impossible to avoid being part of a chain involving some non-negligible harm, even if it occurs many degrees of separation away. Think about the examples of buying eggs or chocolate mentioned above. Therefore, the argument of easy compliance that was valid in the old harms does not apply to the new harms. Due to the degrees of separation, it has become easier than ever to avoid feeling responsible, yet harder than ever to avoid being responsible.

The positive valence also works differently in cases of what we might call “new benefits”, which are the counterpart of new harms. The current abundance of charity projects around the world makes helping an unprecedented number of people easier and cheaper than ever. These new benefits are particularly easy because they are indirect and can occur at several degrees of separation. For example, it is possible to fund simple, cost-effective interventions abroad, such as the distribution of fortified food at a very low cost. In this scenario, the idea that helping is more demanding than doing no harm becomes less evident (Lichtenberg, 2010). Furthermore, as helping becomes easier due to its indirect nature, it becomes crucial to examine our attitudes towards assistance provided at increasingly greater degrees of separation. Therefore, the present investigation also includes positive-valenced conditions dealing with helping instead of harming.

## **Design**

To recapitulate, the goal was to examine the effect of differences ranging from 0 to 4 degrees of separation on the evaluation of helpful or harmful behavior, presented either jointly or separately.

## **Conditions**

The scenarios explored two common contexts in which modern citizens may find themselves involved in causal chains of either causing harm or helping.

Harming conditions: buying and investing.

Helping conditions: buying and donating.

The case of buying, present in both valences, will serve as an illustration [*positive in brackets*]:

0: Person A owns Company Co, which explores [*helps*] employees.

1: Person B is a distributor who buys cheaper[*more expensive*] materials from a Company Co, which explores [*helps*] employees.

2: Person C is a manufacturer who produces cheaper [*more expensive*] products by buying materials from a distributor who sells materials from Company Co, which explores [*helps*] employees.

3: Person D is a retailer who sells cheaper [*more expensive*] products by buying them from a manufacturer. The manufacturer produces cheaper [*more expensive*] products by buying materials from a distributor who sells cheaper [*more expensive*] materials from Company Co, which explores [*helps*] employees.

4: Person E is a consumer who buys a cheaper [*more expensive*] product from a retailer who sells cheaper [*more expensive*] products by buying them from a manufacturer. The manufacturer produces cheaper [*more expensive*] products by buying materials from a distributor who sells cheaper [*more expensive*] materials from Company Co, which explores [*helps*] employees.

Appendix I contains the exact texts. The most important content choices were:

- Third-person was used because the goal was to extract people's normative judgments (Bicchieri et al., 2014).
- All individuals were men to avoid the possibility of different genders being judged differently.
- In every vignette, only the person whose responsibility is under scrutiny is identified by a proper name, such as John or Herman. This was done to avoid the tendency of group action to minimize responsibility (Bandura, 1999). Similarly, the responsibility of enterprises, which may differ, was not explored.
- Degree 0 has no mediator, thus characterizing direct responsibility, but with no physical contact. Thus, even if degrees 0 and 1 contain a direct versus indirect divide, the entire sequence avoids the strong effect of physical contact, enabling a more homogeneous comparison.
- Foreknowledge of harm is necessary for attributing responsibility for indirect harm (Paharia et al. 2009). To avoid any doubt, it was made clear at every level that the person in question was aware of the harm that occurred further down the chain. For instance: "Everyone—including Mark—knows that Clothes & Co. achieves these high profits by subjecting its workers to dangerous and exploitative working conditions." Such foreknowledge is not unrealistic, as most people are aware of new harms, even if they choose to ignore the specifics.
- Given the focus on the conflict between helping or harming others and self-interest, it was made clear at all levels and in all conditions what the cost or advantage was to the individual in question.

## Presentation

Two modes of presentation were used.

Joint Evaluation: Each participant can see all the degrees (0-4) on the same page. For each degree, they must indicate how good or bad the described behavior is on a scale from 0 (morally neutral) to 100 (extremely wrong or good). See below for more information about the scale. Since all conditions appear on the same page, participants can read, compare, and evaluate each degree. They could also reread and change their evaluations as many times as they wanted before clicking the next button. Such a joint presentation encourages a holistic approach to the supply chain. The

flexibility to adjust judgments reduces the impact of anchoring effects. There was just one context for the joint conditions: buying for either help or harm valences. Finally, the same participant evaluated both valences, and the order of appearance of each valence was randomized.

Separate Evaluation: each participant sees only one degree of separation (0, 1, 2, 3, or 4) to avoid benchmarking. Participants see both valences (harm and help) and both contexts (buying and investing for harm and buying and donating for help). However, the degree one sees in the harm block is always different from the degree one sees in the help block. For example, if a participant sees one degree of separation in the help block, they will not see one degree of separation in the harm block. The order of appearance for harm or help, as well as the order of the contexts within each valence, was randomized.

Lastly, a warning page appeared before the presentation of the harm or help block in both joint and separate modes of presentation. The page announced that the valence was going to change (from harming actions to helping actions or vice versa) and that the scale was going to change (from neutral to bad or good).

## **Scales**

Studies on preference reversals (Hsee et al. 1999) — including those involving the preference for indirectness (Paharia et al. 2009) — typically use different evaluation scales. In the separate mode, participants estimate a quantitative value for conditions A and B, whereas in the joint mode, they are asked to choose between A and B. Using different questions is not ideal and requires very conservative analyses. However, since the present study focuses on moral evaluations at each degree rather than preferences, it was possible to use the same type of scale for both the separate and joint modes.

After reading the scenario(s), the participants read the question: How morally wrong [*good*] do you think [character name]'s behavior was? The character's name made clear whose behavior was under evaluation. Using "morally" instead of "ethically" emphasizes an everyday moral context.

Participants then used a unidirectional VAS (visual analogue scale) to indicate how much they considered a given behavior morally wrong or good within a moral score range from 0 (morally neutral) to 100 (extremely morally wrong [*good*]). Since the actions clearly involved either a harm or a help done, it would make no sense to provide a full bidirectional scale. The choice to use "good" instead of "right" (the immediate opposite of "wrong") was motivated by the tendency to see helping as morally good but supererogatory, rather than obligatory. Thus, the axiological "good" was preferable to the normative "right." Finally, the 0-100 scale is better suited for evaluating five degrees of separation than a Likert scale, which would lead to statistical compression.

## **Accountability**

In addition to the VAS scales, participants in the harm-separate condition were asked to answer an accountability question.<sup>1</sup> They read:

In your opinion, should [character name] be held accountable for this action?  
( ) Yes, legally accountable (e.g., fines, lawsuits, or criminal charges).

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<sup>1</sup> The question of accountability does not naturally apply to help cases. That is why no questions were asked in this situation. The joint mode did not have a question about accountability either. Since all the cases appeared on the same page, adding yet another question at each level would have been too confusing.

- ( ) Yes, socially accountable (e.g., public criticism, boycotts, but no legal action).  
( ) No.

This question provides a more direct measure of the moral decoupling point (Eriksson & Svensson 2016). In this context, it refers to the degree of separation at which responsibility is canceled out, if any. The options also capture the distinction between legal and social responsibility. Previous research has found significant differences in judgments of punishment versus blame, to the extent that some theories attribute them to different processes (Kneer and Machery 2019).

## Preregistration

Preregistration, data, and materials are available at: <https://osf.io/6qcdf/>.

Participants were recruited via Prolific. The study was conducted in 05.06.2026, using the online survey software SoSci Survey (Leiner, 2026). After quick responses were rejected, a total of 179 participants ( $M_{\text{age}} = 36.5$  years,  $SD = 12.25$ ; 78 female, 98 male, 3 non-binary/other) were used for the analysis.<sup>2</sup>

## Expectations

Harm-separate:

The constitutive question of the study was whether degrees of separation affect the preference for indirect harm. The general expectation was that each degree would ease moral judgments. Previous studies provided strong enough evidence to suggest that the difference between 0 (direct) and 1 (indirect) would be significant, at least in separate conditions (Royzman & Baron, 2002; Paharia et al. 2009). There is no data on further degrees, but the differences should be smaller. An ANOVA will test for relative differences between all degrees (0–4). Post hoc pairwise comparisons with Bonferroni correction will compare each degree to the others. Then a linear regression analysis will reveal the rate at which evaluations decay. Additionally, the accountability question will determine at what degree of separation indirect harm ceases to be perceived as legally and socially condemnable (Eriksson & Svensson 2016).

Harm-joint:

The same analyses will be performed regarding the joint mode of evaluation. Additionally, a mixed two-way ANOVA testing mode (joint or separate) and five levels of degree (0-4) will reveal whether there is an interaction between these factors. The preference reversal observed in studies comparing direct and indirect actions suggests an interaction effect (Hsee et al. 1999, Hsee et al., 2005). Furthermore, economists hypothesize that a holistic view of the supply chain could prevent moral disengagement (Ashby et al. 2012). If so, judgments across degrees should differ less in the joint mode of presentation. Comparing the regressions between the two modes of presentation may provide insight into this question.

Help-separate and help-Joint

The positive valence (helping) will undergo the same analyses in both modes of presentation, separate and joint. Additionally, a mixed two-way ANOVA testing valence (harm or help) and degrees of separation (0–4) will reveal whether an interaction effect exists between these factors.

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<sup>2</sup> An *a priori* power analysis conducted using G\*Power (Version 3.1; Faul et al., 2009) indicated that minimum sample sizes of  $N = 50$  per condition for the separate evaluation design and  $N = 30$  per condition for the joint evaluation design yielded adequate statistical power ( $1 - \beta \geq .80$ ,  $\alpha = .05$ ) to detect medium effect sizes ( $f = 0.25$  / Cohen's  $d = 0.50$ ). These sample sizes align with similar empirical work (e.g., Paharia et al., 2009), which observed medium-to-large effect sizes ranging from  $d = 0.50$  to  $d = 0.71$ .)

Studies show that people attribute stronger intentions and responsibility to agents performing wrong actions (Knobe 2003; Cova, 2016; Feltz, 2007), suggesting an interaction effect will be found.

## Results

The descriptive mapping shows the means and standard errors for each of the five degrees in the joint (solid line) and separated (dashed line) modes of evaluation, as well as for the harm (orange) and help (green) valences (See the annex II for the descriptive statistics table).

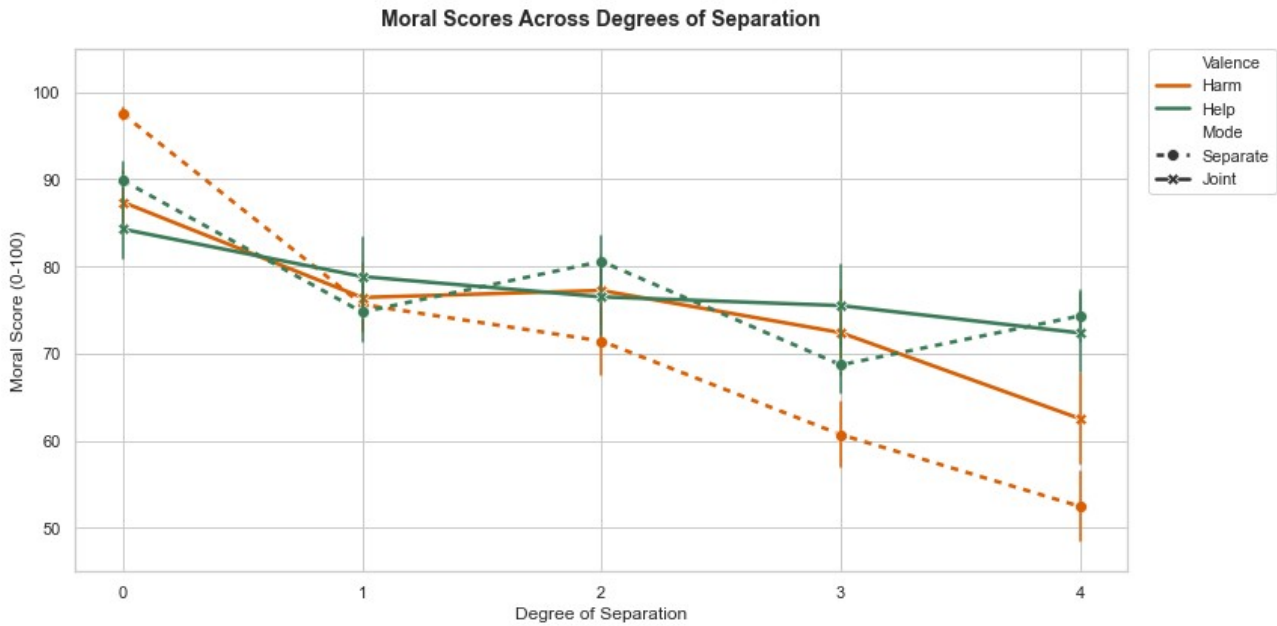


Figure 1: Mean moral scores and their respective confidence intervals across degrees of separation for all conditions.

An initial visual inspection reveals a general trend. The differences in harm are more extreme than those in help. The differences in separate are more extreme than those in joint. Help-joint is the smoothest, approaching a flat line. In contrast, the harm-separate line starts higher than all the others, experiences the steepest decline through each degree, and ends at the lowest point (roughly 50%, or neutral).

## Statistical Analyses

### Harm-separate

A 2 (Context: Buying vs. Investing) x 5 (Degree of Separation: 0 to 4) two-way ANOVA revealed a highly significant main effect of Degree of Separation ( $F(4, 290) = 26.75, p < .001, \eta^2_p = 0.26$ ). In contrast, neither the main effect of Context ( $F(1, 290) = 0.53, p = 0.47, \eta^2_p = 0.001$ ) nor the Context x Degree interaction ( $F(4, 290) = 2.15, p = 0.074, \eta^2_p = 0.02$ ) reached statistical significance.<sup>3</sup>

<sup>3</sup> In our pre-registration, we specified an LMM to evaluate random variance across evaluation contexts. However, fitting this structure yielded convergence warnings (singular fit), indicating that the random-effect variance components were negligible and that additional context-level noise was absent. Thus, we only report the pre-registered ANOVAs.

Given the pronounced main effect of distance, post-hoc pairwise comparisons were conducted to further examine the precise trajectory of moral decay across degrees.

As expected, the transition from direct action (Degree 0) to indirect action (Degree 1) marked a highly significant drop in blame ( $t(118) = 6.84$ ,  $g = 1.24$ ,  $p_{\text{corr}} < .001$ ). After this, cumulative distance produced significant incremental decay when spanning two or more degrees of separation, as evidenced by significant differences between Degree 1 and Degree 3 ( $t(118) = 3.06$ ,  $g = 0.55$ ,  $p_{\text{corr}} = .027$ ), Degree 2 and Degree 4 ( $t(118) = 3.40$ ,  $g = 0.61$ ,  $p_{\text{corr}} = .009$ ), and Degree 1 and Degree 4 ( $t(118) = 4.62$ ,  $g = 0.83$ ,  $p_{\text{corr}} < .001$ ). Taken together, these results reveal that the preference for indirectness occurs in harm-separate condition. Moreover, the degrees of indirectness matter but operates along a continuous spectrum of cumulative decay rather than discrete categorical plateaus. A linear regression shed more light on how the progression works.

An OLS linear regression was conducted predicting moral blame from Degree of Separation (with Degree 0 as the reference category) while controlling for supply chain Context (Buying vs. Investment).<sup>4</sup> The first column of table 1 shows the results.

| Predictor               | Harm-Separate<br>(B)                  | Help-Separate<br>(B)       | Harm-Joint<br>(B)                      | Help-Joint<br>(B)                   |
|-------------------------|---------------------------------------|----------------------------|----------------------------------------|-------------------------------------|
| Intercept<br>(Degree 0) | <b>96.38***</b><br>(3.65)             | <b>88.95***</b><br>(3.25)  | <b>87.37***</b><br>(4.30)              | <b>84.27***</b><br>(4.35)           |
| Degree 1                | <b>-21.83***</b><br>(4.71) 74         | <b>-15.10***</b><br>(4.20) | <b>-10.93<sup>^</sup></b><br>(6.08) 76 | <b>-5.43</b><br>(6.15)              |
| Degree 2                | <b>-26.03***</b><br>(4.71) 70         | <b>-9.28*</b><br>(4.20)    | <b>-10.13</b><br>(6.08) 77             | <b>-7.77</b><br>(6.15)              |
| Degree 3                | <b>-36.77***</b><br>(4.71) 59         | <b>-21.15***</b><br>(4.20) | <b>-14.97*</b><br>(6.08) 72            | <b>-8.77</b><br>(6.15)              |
| Degree 4                | <b>-45.00***</b><br>(4.71) 51<br>=350 | <b>-15.53***</b><br>(4.20) | <b>-24.83***</b><br>(6.08) 62<br>= 374 | <b>-11.90<sup>^</sup></b><br>(6.15) |

Table 1: Comparison of OLS regressions across all the conditions.

*Note.* Unstandardized B regression coefficients are displayed with standard errors in parentheses below each value. Reference category for Degree of Separation is Degree 0 (Direct Action).

<sup>^</sup> $p < .10$  \* $p < .05$ .  $p < .01$ . \*\*\* $p < .001$

<sup>4</sup> The different contexts (buying or investing) were not statistically different from one another ( $\beta = 2.17$ ,  $t = 0.73$ ,  $p = .467$ ), at least when all degrees were taken into account. The case of degree 4 will be explored below.

Baseline blame for direct harm (Degree 0) was near ceiling ( $B = 96.38$ ,  $SE = 3.65$ ,  $p < .001$ ). Introducing a single layer of separation (Degree 1) triggered an immediate 21.83-point drop in condemnation ( $B = -21.83$ ,  $p < .001$ ). Blame continued to erode across subsequent links, dropping by 26.03 points at Degree 2 ( $B = -26.03$ ,  $p < .001$ ), 36.77 points at Degree 3 ( $B = -36.77$ ,  $p < .001$ ), and culminating in a cumulative 45.00-point reduction at Degree 4 ( $B = -45.00$ ,  $p < .001$ ). At this final node, predicted moral blame reached 51.38%, representing a point of moral absolution on a neutral-to-blameworthy scale.

## Accountability

To explore the categorical nature of the decay in moral judgments, participants' assignments of accountability (Legal, Social, or None) were analyzed across the five degrees of separation. A Chi-Square Test of Independence revealed a highly significant shift in how accountability was assigned across distances for both contexts ( $\chi^2(8) = 95.32$ ,  $p < .001$ ).

The graphics (figure 2) show the percentage of participants who said the behavior in question should be punished legally, socially, or not at all. The two contexts, buying and investing, are presented separately. Each bar represents a degree.

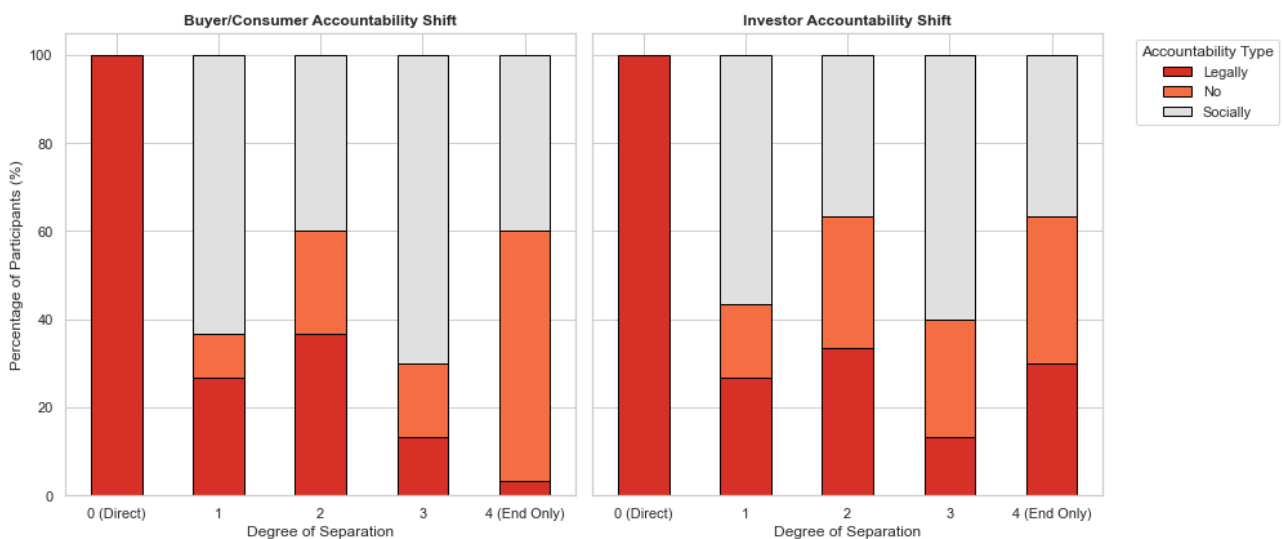


Figure 2: Percentage-based responses to the accountability question in two different contexts.

The bars show a perfect consensus that the direct action should be legally punished. Once one degree of separation is established, legal accountability drops to approximately 30% in both contexts, which confirms the preference for indirectness. By contrast, social blame proves to be more lasting, with a significant percentage of participants believing that individuals with two to four degrees of separation should still be socially reprimanded.

A comparison between degrees four reveals the most notable difference between the contexts. In the case of buying, the majority believes that not even social condemnation is applicable to someone at

4 degrees of separation (orange block, 56%). Interestingly, this is the position of the final consumer, a role that all participants find themselves in. In contrast, most people (~65%) believe that investors at four degrees of separation should be punished, either legally or socially.

### **Harm-joint**

To test whether the evaluation mode altered the trajectory of moral decay across distance within the harm condition, a 2 (Mode: Separate vs. Joint) x 5 (Degree of Separation: 0 to 4) mixed ANOVA was conducted. The analysis revealed a statistically significant Evaluation Mode x Degree interaction,  $F(4, 440) = 2.45$ ,  $p = .046$ ,  $\eta^2_p = 0.02$ , confirming that the rate at which distance blunts moral condemnation depends fundamentally on presentation mode. To isolate how a joint evaluation alters this pattern, condition harm-joint was examined alone.

Post-hoc pairwise comparisons with Bonferroni correction found that only the difference between 0 and 4 degrees is statistically significant ( $t(58) = 3.96$ ,  $p_{\text{corr.}} = .002$ ). Unlike separate evaluation, the initial transition from direct to indirect harm (Degree 0 vs. Degree 1) was not statistically significant ( $t(58) = 2.07$ ,  $p_{\text{corr.}} = .428$ ). Moreover, to model how moral decay progresses across distance, an OLS linear regression was conducted with Degree of Separation entered as a categorical predictor (Degree 0 as the reference group).

Table 1 facilitates the comparison between both harm conditions. Direct harm yielded a near-ceiling intercept under separate evaluation ( $B = 96.38$ ,  $p < .001$ ), which was noticeably diluted under joint evaluation ( $B = 87.37$ ,  $p < .001$ ). Crucially, the moral score fall at Degree 4 was nearly halved in the joint condition relative to the separate condition ( $B_{\text{joint}} = -24.83$ ,  $p < .001$  vs.  $B_{\text{separate}} = -45.00$ ,  $p < .001$ ). While blame under separate evaluation decayed to a virtually neutral threshold at Degree 4 (predicted score of 51.38), joint evaluation maintained substantial moral liability across the entire chain, ending at a predicted score of 62.54. Thus, a joint presentation incentivizes a more resilient distribution of responsibility throughout the entire chain, including the final consumer.

### **Help-Separate**

A 2 (Context: Buying vs. Donating) x 5 (Degree of Separation: 0 to 4) two-way ANOVA revealed a highly significant main effect of Degree of Separation ( $F(4, 290) = 7.38$ ,  $p < .001$ ,  $\eta^2_p = 0.09$ ). In contrast, neither the main effect of Context ( $F(1, 290) = 0.46$ ,  $p = 0.4$ ,  $\eta^2_p = 0.001$ ) nor the Context x Degree interaction ( $F(4, 290) = 2.02$ ,  $p = 0.09$ ,  $\eta^2_p = 0.02$ ) reached statistical significance. Given the pronounced main effect of distance, post-hoc pairwise comparisons were conducted to further examine the precise trajectory of moral decay across degrees.

The difference between direct and indirect (degrees 0 and 1) is once again significant in a separate evaluation ( $t(118) = 3.72$ ,  $g = 0.68$ ,  $p = .002$ ), however the effect size is bigger in the harm valence ( $g = 1.24$ ). Following this initial reduction, adjacent intermediate steps showed no significant pairwise shifts from degree 1 to 2, 1 to 3, and 1 to 4 (all  $p_{\text{corr.}} = 1.000$ ), demonstrating that praise rapidly plateaus once direct agency is removed. An OLS linear regression with Degree of Separation entered as a categorical predictor (Degree 0 as the reference group) sheds more light on how the progression works.

Once again, table 1 allows for a comparison between the progressions. Help-separate exhibits the preference for indirectness shown in the big difference between degrees 0 and 1, like harm-separate. Even though, help-separate progression is closer to harm-joint. Baseline laudability for direct helping behavior ( $B = 88.95$ ) starts near the baseline of joint harm ( $B = 87.37$ ), which is way lower than the near-ceiling baseline of separate harm ( $B = 96.38$ ). Crucially, total erosion at the terminal node (Degree 4) for help-separate is only -15.53 points—considerably less than the -24.83-point drop observed in joint harm, and drastically smaller than the -45.00-point drop in separate harm. Thus, valence exhibits asymmetry. Even when evaluated separately, praise for indirect positive actions proves far more resilient to distance-based erosion than blame for indirect harm.

## Help-Joint

An omnibus ANOVA revealed no statistically significant differences across degrees of separation in the joint help condition ( $F(4, 145) = 1.04, p = .387, \eta^2_p = 0.02$ ). Consistent with this, the OLS regression model (Table 1) confirmed that moving from direct agency (Degree 0) to four layers of supply chain separation (Degree 4) produced a negligible -11.90-point shift in praise that failed to reach statistical significance ( $B = -11.90, SE = 6.15, p = .055$ ). While joint evaluation systematically flattened decay trajectories across both valences, combining joint presentation with help valence produced the most resilient condition in the entire study. In help-joint condition, indirect help, even at four degrees of separation, is essentially as praiseworthy as direct help.

## Degree vs. Mode vs. Valence

Finally, to compare the relative strength of Evaluation Mode and Valence in moderating moral decay across degrees of separation, we examined their respective two-way interaction effect sizes. First of all, degree alone accounts for nearly a large effect ( $F(4, 880) = 32.08, \eta^2_p = .127, p < .001$ ). Secondly, while both Mode  $\times$  Degree ( $F(4, 880) = 2.65, p = .032, \eta^2_p = .012$ ) and Valence  $\times$  Degree ( $F(4, 880) = 6.04, p < .001, \eta^2_p = .027$ ) reached statistical significance, valence exerted more than double the moderating influence of evaluation mode. Finally, the non-significant three-way interaction ( $p = .428, \eta^2_p = .004$ ) further indicates that these two moderating mechanisms operate independently across psychological distance.

## Discussion

The results provide initial evidence for a complex picture of the role of degrees of separation in moral evaluations. The importance of distance is clear ( $\eta^2_p = .127, p < .001$ ). The two independent variables—the mode of evaluation (separate versus joint) and valence (harm versus help)—exert a small moderating influence ( $p = .032$  and  $p < .001$ ). In general, separate evaluation and/or negative valence increase the differences in moral evaluations across degrees, whereas joint evaluation and/or positive valence decrease them.

The effects of each variable are independent of one another (non-significant three-way interaction,  $p = .43$ ) and either reinforce or diminish when combined. Combinations of variables that impact the results in the same direction present the most extreme cases. Harm-separate generates the greatest

differences, while help-joint generates the least. Combinations with an opposite direction of impact mitigate the effects. For example, harm-joint is milder than harm-separate and help-separate, but stronger than help-joint.

Valence exerts more than double the moderating influence of evaluation mode ( $\eta^2_p = .027$  vs.  $.012$ ). Thus, helping those who help others, even with several degrees of separation, remains highly laudable, whereas moral blame decreases more radically with each degree of separation from the harm until it vanishes by degree 4 of the separate condition.

In line with studies on the preference for indirectness that adopted a binary approach to distinguish between direct and indirect actions (Royzman & Baron, 2002; Waldmann & Dieterich, 2007; Cushman, et al., 2006), we found a significant difference between degrees 0 (direct) and 1 (indirect), but only in separate evaluations: harm-separate ( $p < .001$ ) and help-separate ( $p = .002$ ). In contrast, the difference between degrees 0 and 1 was not significant in any joint condition. This confirms that prompting a reflective comparison mitigates the effects of the preference for indirectness (Hsee et al., 1999, Paharia et al. 2009). Lastly, the significant effects of degrees of separation suggests that a binary approach could overlook important aspects.

The regression models for moral scores further enrich our understanding of how degrees of indirectness affect outcomes. The progression of effects reveal a gradual cumulative effect better modeled as a dimensional spectrum than distinct categories or steps, with steeper curves under separate evaluation and smoother ones under a joint evaluation. Specific insights will be provided by individual analyses of each condition.

### **Harm-separate**

The initial judgment in the harm-separate condition is so harsh that almost hits the ceiling at 96 points. In contrast, the initial judgment in the joint evaluation was 87 points. The extreme score in the separate mode aligns with the finding that evaluations made in isolation elicit affective judgments based on high-emotion attributes (Hsee et al., 2005; Hsee et al., 1999; Bazerman et al., 1998). Under this assumption, the mere transition from direct to indirect action arouses significantly fewer emotions, resulting in a 22-point decrease. However, it is important to note that indirect action at degree 1 is still highly blameworthy, with a moral score of around 75. After that, adding each mediator eased the judgments even more, with significant differences between every other degree ( $p < .001$  for both degree 1 vs. 3 and degree 2 vs. 4). However, it was only at degree 4 that the moral score fell within the range of mild moral condemnation to moral neutrality. The harm-separate condition was the only one to reach this boundary at 51. The meaning of this result must be discussed in light of the accountability question.

The harm-separate condition included a categorical question that asked whether the agent should be held legally or socially accountable, or not at all. The aim was to identify the moral decoupling point—the degree of separation at which people stop blaming an agent involved in a supply chain where there is harm (Eriksson & Svensson 2016). The categorical answers are consistent with the moral scores. All participants thought that agents at degree 0 should be legally punished, which aligns with the 96-point moral blame score.

The 22-point drop in score between degree 0 and degree 1 aligns with the transition from legal accountability at degree 0 to social accountability from degrees 1 through 3. Legal accountability clearly involves punishment. Social accountability is closer to blame. In this case, the distinction between direct harm, which deserves legal punishment, and indirect harm,

which deserves blame, may be related to the asymmetric findings that support the hypothesis that punishment and blame stem from different processes (Kneer and Machery 2019, but see Cushman 2008). According to this hypothesis, the basis for determining punishment is a *partial direct effect* on the outcome. If this is the case in the present study, participants do not view degrees 1 to 3 as contributing to the direct harmful effect and thus think that they only receive social blame.

Finally, most people believe that agents at four degrees of separation are not accountable at all, which aligns with the neutral score of 51 points of moral blame. These results indicate that degree 4 of harm-separate reached the moral decoupling point where one ceases to be responsible for the harm down the chain. However, this conclusion requires further analysis since accountability at degree 4 was the only situation in which the different contexts (buying vs. investing) made a difference.

As figure 2 shows, for buyers, no accountability won by a landslide (68% vs. 4%), but among investors, a substantial group of participants still demanded some sort of accountability (64%), with legal accountability even experiencing a resurgence at this final stage (36%). The difference is consistent with the mean scores. In the investment context, degree 4 of harm-separate received a score of 61.56, while in the buying context, the score was 43.36, which is well below the midpoint. Therefore, moral decoupling does not occur in the case of investors.

In light of the *partial direct effect* that justifies punishment judgments (Kneer and Machery 2019), one might suppose that investors are more likely to be perceived as directly funding harm. In contrast, consumers purchase products that have already been produced, so they may not be seen as direct causes. In moral vocabulary, the distinction can be put in terms of investors being seen as causing harm, while consumers are seen as only enabling it. The relevant difference is that an enabler is not part of the initial action (Barry & Øverland, 2012; Barry et al., 2014). However, this explanation is insufficient since other agents in the buying context such as retailers at degree 3 also purchase existing products and even so they are still condemned,.

A notable difference is that participants are much more likely to be consumers than investors (62% of Americans and 58% of Britons invest). Thus, being part of a harmful chain becomes permissible exactly when the scenario describes the position participants find themselves in. This self-serving moral decoupling point is not surprising. Eriksson & Svensson (2016) find a similar behavior in managers who want to avoid responsibility for what happens down the supply chains. Thus, instead of a fixed decoupling point at degree 4 or at the end-consumer level, there seems to be a moving decoupling point that we tend to push away from the position we are in.

In the present study, the self-serving moral disengagement does not seem to characterize motivated reasoning. After all, in the joint evaluation, which encourages reflection, people still condemn the final consumer. Thus, absolving oneself in the separate mode seems to reflect a mitigation of the emotional response, perhaps because this behavior is so ordinary.

## **Harm-Joint**

A comparison of harm-separate and harm-joint revealed a significant interaction effect between evaluation method and degree ( $F(4, 440) = 2.45, p = .046, \eta^2_p = .022$ ). Regression models for each mode showcase informative differences (see Table 01). The moral blame score in harm-joint begins lower at 87 and ends higher at 63 (compared to 96 and 51, respectively, in the separate mode). Note that harm-joint only tested the buying context, which had an even lower score of 43 at degree 4 in harm-separate, as seen above. Thus, the joint evaluation was sufficient to counteract the exculpation

of the final end consumers observed in the separate evaluation. This suggests that encouraging a reflective comparison of the entire supply chain reduces the risk of self-serving moral disengagement that exempts end consumers from blame, which is in line with a holistic view of the supply chain (Ashby et al., 2012). A comparative examination of degree 0 provides additional insight.

The lower initial blame score for joint harm (87 instead of 96) indicates that the stronger blame assigned to final consumers in degree 4 is accompanied by a less severe condemnation of direct agents in degree 0. This inverse relationship is informative. Part of the self-serving moral disengagement found in the separate evaluation at degree 4 seems to be due to responsibility diffusion (Bandura, 1999; Bandura et al., 1996). Specifically, in the separate evaluation, the direct agents at degree 0 receive almost all the responsibility. This strong judgment seems sufficient to absolve those at the other end of the chain. In contrast, when some responsibility is attributed to final consumers, less responsibility is allocated to other agents in the chain, particularly those directly involved. Thus, the smoother curve of progression, delimited by a lower initial score and a higher final score, indicates that a joint evaluation promotes a fairer distribution of responsibility throughout the causal chain.

The distribution of responsibility in the joint evaluation of a supply chain sheds light on another counterintuitive aspect of new harms. Regarding traditional harms, Bandura (1999) shows that vilifying victims is an effective way to achieve moral disengagement, which allows people to harm others. The rationale is intuitive. Once victims are vilified, people become more willing to treat them as if they deserve the harm. However, the above comparison suggests another mechanism. In the context of new harms, vilifying those who directly commit the harm also contributes to moral disengagement. This is counterintuitive because being outraged at those who commit the harm, like the participants in harm-separate degree 0, feels like moral engagement. However, this feeling is deceptive. Placing all the blame on the direct agent allows the other indirect agents in the chain to disengage from their moral responsibility and simply pursue their self-interest. As economic games demonstrate, we avoid acquiring information about the harm involved in our actions in order to act in a self-interested manner without feeling guilty (Dana et al., 2003). The present results indicate that we can also do so by attributing all the blame to those closer to the harm. Once again, the new harms are tricky enough to require a counterintuitive type of moral awareness from us.<sup>5</sup>

## Help-separate

The difference between degrees of separation in the help-separate condition was statistically significant ( $F(4, 290) = 7.38, p < .001, \eta^2_p = 0.09$ ). According to the regression model, evaluations begin with a moral praise score of 89 points for direct help. This score is lower than the 96 points for harm-separate and similar to the 88 points for harm-joint and the 84 points for help-joint. If separate evaluations are more emotional, direct helping did not generate as strong an emotional response as direct harm. This is consistent with the tendency for negative valence to capture more attention in morality. People tend to be more negatively affected by harmful actions than positively affected by prosocial behavior (Baumeister et al., 2001).

Similar to harm-separate, the difference between degrees 0 and 1 (direct vs. indirect action) in help-separate was statistically significant ( $t(118) = 3.72, g = 0.68, p = .002$ ), though the effect size was smaller. However, the similarities end here. After degree one, help-separate becomes much more

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<sup>5</sup> One might fear that a fairer distribution of responsibility could reduce the pressure for change in enterprises more involved in the supply chain. However, this does not seem to be the case since degrees 1 to 3 receive higher blame in the joint evaluation than in a separate evaluation.

stable than harm-separate. There was no significant difference in the degrees of indirectness. In fact, there was even some oscillation. Descriptively, degree 2 received more praise than degree 1, and degree 4 received more praise than degree 3. The flat, oscillating curve suggests that in a positive valence, there are no degrees of indirectness. A binary opposition between direct and indirect help is sufficient to capture the difference in evaluations. This does not diminish the importance of the analysis, of course. Taking degrees of separation into account was necessary to reach the nonobvious conclusion that moral praise does not decay with distance.

We have seen that consumers who are four degrees of separation away are no longer held accountable for harm further down the causal chain. In contrast, an ethically conscious consumer or donor four degrees away from the provided help still deserves praise for the distant benefit (73). In short, consumers are praised for ethical consumption and not blamed for unethical consumption when evaluated separately. While asymmetries between positive and negative valences are common, the direction of this asymmetry is rather unexpected.

Previous research shows that people are more willing to attribute intention and responsibility to harmful actions than to beneficial ones (Knobe, 2003; Cova, 2016; Feltz, 2007). This is in line with the harsher judgments seen at degree 0 of harm-separate. Nevertheless, an analysis of degrees of indirectness reveals a nuance. In the case of indirect helpful behavior, such as ethical consumption and donations, distant agents remain seen as highly responsible for positive outcomes, even with four degrees of separation. Thus, the willingness to attribute intention and responsibility to indirect actions may exhibit reverse asymmetry, favoring positive behavior over negative behavior.

The resilience found in moral praise is another way in which our positive-valenced morals seem better equipped to deal with new harms than negative ones. As we have seen, it is easier to refrain from traditional harms than to perform positive actions. For instance, it is simpler to avoid harming someone through physical violence than to help them directly. New harms reverse this scenario. It has become difficult to avoid harming people since many of our everyday actions result in distant harm. In contrast, a vast network of effective charities has made helping others from afar easier than ever. These are indirect acts, usually with many degrees of separation from the assistance provided. This aligns well with the observed resilience of moral praise in separate evaluations. Thus, even in emotional judgments, people will be morally praised and feel proud for helping indirectly, with many degrees of separation from the benefits they cause.

The resilience of moral praise for indirect actions by emotional judgments aligns with the warm glow model of altruism. According to this model, people are emotionally motivated to donate because of the warm feeling they receive as a reward (Andreoni, 1990). However, one should not assume that such praise, even if it is emotional, will be enough to motivate behavior. For one thing, even though people want businesses to be ethical, they do not act accordingly in that they are not willing to pay more for ethical consumption (Auger and Devinney 2007; Carrington et al. 2010; Öberseder et al. 2011).

## **Help-joint**

Help-joint was the only condition in which an ANOVA revealed no statistically significant differences across degrees of separation ( $F(4, 145) = 1.04, p = .387, \eta^2_p = 0.02$ ). Accordingly, the regression model yielded the flattest curve. There was no significant difference between degrees 0 and 1, i.e., direct or indirect helpful behavior. This is similar to the other joint condition (harm-joint) and different from the separate conditions (harm-separate and help-separate).

Once again, the results in a joint condition are more extreme than in the separate one. In the case of joint-harm, it sounds plausible that the reflective condition leads to a fairer distribution of responsibility throughout the chain. However, in the case of help, it seems surprising that a reflected comparison attributes no statistically significant difference in praising someone who directly helps people in need and someone who is four degrees away from the actual help. One may even wonder whether such indiscriminate attribution of moral praise is justified. Maybe participants are simply rationalizing their way into being virtuous without direct engagement. After all, reflection also fails. However, this does not seem to be the case.

First, note that the results do not indicate a perfectly flat line. While the results are extreme, all the tendencies observed in other conditions are present. The evaluation starts high. The drop from degree 0 to 1 is slightly larger than the others. There is also a marginal effect between degrees 0 and 4 ( $p = .55$ ). Thus, we are presented with a more extreme version of what we observed in the separate condition, which seemed plausible. More importantly, this extreme version reinforces the idea that our praise judgments are suitable for the new benefits, which are the counterpart of the new harms. Reflection increases moral praise for indirect help rather than diminishing it. Therefore, the warm glow experienced when one donates does not have to be an emotional reaction. This is another case in which reflection may be more effective in prompting prosocial behavior.

Given the persistence of moral praise in helping behavior, future research should examine its practical effects. Although people want enterprises to guarantee ethical behavior throughout their supply chains, they are not willing to pay more for it. It remains to be seen whether jointly evaluating a supply chain that produces more expensive products to support workers would align consumer behavior with their moral demands.

## **Conclusion**

Unlike the usual treatment of our attitudes toward direct and indirect moral actions as a binary question, this study takes a step forward by treating indirectness as a matter of degrees. This approach can capture how adding increasingly more mediators affects moral blame or praise. This is important in a globalized market in which complex supply chains create many steps of distance between harmful actions at one end of the chain and self-interested advantages at the other. Similarly, a complex network of charities in a globalized third-sector facilitates assistance with many degrees of separation.

Degrees of separation has the largest effect. The interaction with valence (harm x help) and mode (separate x joint) is also significant. Separate evaluations reveal a significant difference between degrees 0 (direct) and 1 (indirect) of separation. Furthermore, subsequent degrees of indirectness have a cumulative effect, which is stronger in a harm chain than in a help chain. In harm-separate, the accumulation is sufficient to neutralize moral blame for a harmful action at four degrees of separation. This stronger distance effect increases the risk of moral disengagement by displacing all responsibility to distant agents down the chain. Current supply chains exploit this by assuming responsibility only for the behavior of the closest suppliers.

In contrast, a holistic approach that considers all degrees together erases the significance of degrees 0 and 1. Furthermore, the distribution of blame or praise throughout the entire chain becomes more equitable. This is particularly true in the case of help. In help-joint, the moral praise given to direct help is not statistically different from the praise given to an agent four degrees of separation away. The resilience of moral praise aligns with the way current charities operate, relying on donations

from distant donors. The results suggest that these donors both feel and reflectively recognize the praise for their prosocial behavior.

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## **Appendix I – Materials**

### **NEGATIVE**

#### **Investing**

- 0) Steve, the CEO of Clothes & Co., maximizes profits by forcing his workers to work in dangerous and exploitative conditions.
- 1) Howard invested a significant sum in Clothes & Co. because the company offers unusually high annual returns. Everyone—including Howard—knows that Clothes & Co. achieves these high profits by subjecting its workers to dangerous and exploitative working conditions.
- 2) Herman invested a significant sum in a Mutual Fund because it offers unusually high annual returns. The fund achieves these high returns because a major part of its portfolio includes Clothes & Co. Everyone—including Herman—knows that Clothes & Co. achieves these high profits by subjecting its workers to dangerous and exploitative working conditions.
- 3) Mark followed his financial advisor's recommendation and invested a significant sum in a Mutual Fund that offers unusually high annual returns. The fund achieves these high returns because a major part of its portfolio includes Clothes & Co. Everyone—including Mark—knows that Clothes & Co. achieves these high profits by subjecting its workers to dangerous and exploitative working conditions.
- 4) Richard followed in his father's footsteps by investing a substantial sum in a Mutual Fund that offers unusually high annual returns. His father received the investment tip from his financial advisor. The fund achieves these high returns because a major part of its portfolio includes Clothes & Co. Everyone—including Richard—knows that Clothes & Co. achieves these high profits by subjecting its workers to dangerous and exploitative working conditions.

#### **Buying**

- 0) Sean reduces the prices of the precious stones he sells by enforcing dangerous and exploitative working conditions for his miners.
- 1) John exports precious stones. He buys them at an unusually low price directly from an overseas mine. John is aware that the lower prices are possible through the dangerous and exploitative working conditions imposed on the miners.
- 2) Gary produces jewelry with lower-priced precious stones that he acquires from an import company. The importer buys the stones at an unusually low price directly from an overseas mine. Gary is aware that the stones are cheaper because the mine enforces dangerous and exploitative working conditions on its miners.
- 3) Garret owns a jewelry store where he sells competitively priced jewelry produced by a local producer. The producer manufactures the jewelry using lower-priced precious stones acquired from an importer. The importer buys the stones at an unusually low price directly from an overseas mine. Garret is aware that the lower prices are possible through the dangerous and exploitative working conditions imposed on the miners.
- 4) Dale bought his wedding rings at a traditional jewelry store at a very competitive price. The store sells competitively priced jewelry produced by a local producer. The producer manufactures the jewelry using lower-priced precious stones acquired from an importer. The importer buys the stones at an unusually low price directly from an overseas mine. Dale is aware that the lower prices are possible through the dangerous and exploitative working conditions imposed on the miners.

### **POSITIVE**

#### **Donations**

- 0) Robert runs a charity that provides nutrient supplementation to malnourished children.

- 1) William donates to a charity. The charity provides nutrient supplementation to malnourished children.
  - 2) Howard accepted his colleague's donation request. The colleague is collecting donations for a charity. The charity provides nutrient supplementation to malnourished children.
  - 3) Harry supported his colleagues' request to their employer. The colleague wants their employer to match a donation to a charity. The charity provides nutrient supplementation to malnourished children.
  - 4) Kirk endorsed his wife's decision. She will support her colleagues' request to their employer. The colleague wants their employer to match a donation to a charity. The charity provides nutrient supplementation to malnourished children.
- 0) Arthur, the CEO of Fair Mining & Co., maximizes his workers' well-being by providing exceptionally safe working conditions and thriving wages for his miners, even if it increases the cost of his product.
- 1) Stewart distributes cobalt from FAIR Mining & Co. He informs all of his customers that the cobalt is more expensive because the company maximizes its workers' well-being by providing exceptionally safe working conditions and thriving wages.
  - 2) Trent, the CEO of Tech Electronics, decided to purchase more expensive cobalt from a third-party distributor who sources it from FAIR Mining & Co. Trent knows that the cobalt costs more because FAIR Mining & Co. maximizes its workers' well-being by providing exceptionally safe working conditions and thriving wages.
  - 3) Walter owns a retail store where he sells Tech Electronics' higher-priced devices. They cost more because they use expensive cobalt sourced from a third-party distributor, who buys it from FAIR Mining & Co. Walter knows that the company maximizes its workers' well-being by providing exceptionally safe working conditions and thriving wages.
  - 4) Frederick bought his computer at Walter's retail store. He sells Tech Electronics' higher-priced devices. They cost more because they use expensive cobalt sourced from a third-party distributor, who buys it from FAIR Mining & Co. Frederick knows that the company maximizes its workers' well-being by providing exceptionally safe working conditions and thriving wages.

## Appendix II

### Descriptive Statistics for Moral Scores across Evaluation Mode, Valence, and Degree of Separation

| Mode  | Valence | Degree | N  | Mean  | SD    | SE   |
|-------|---------|--------|----|-------|-------|------|
| Joint | Harm    | 0      | 30 | 87.37 | 19.70 | 3.60 |
|       |         | 1      | 30 | 76.43 | 21.16 | 3.86 |
|       |         | 2      | 30 | 77.23 | 20.81 | 3.80 |
|       |         | 3      | 30 | 72.40 | 26.71 | 4.88 |
|       |         | 4      | 30 | 62.53 | 28.18 | 5.14 |

| <b>Mode</b>     | <b>Valence</b> | <b>Degree</b> | <b>N</b> | <b>Mean</b> | <b>SD</b> | <b>SE</b> |
|-----------------|----------------|---------------|----------|-------------|-----------|-----------|
|                 | <b>Help</b>    | 0             | 30       | 84.27       | 18.95     | 3.46      |
|                 |                | 1             | 30       | 78.83       | 24.14     | 4.41      |
|                 |                | 2             | 30       | 76.50       | 24.82     | 4.53      |
|                 |                | 3             | 30       | 75.50       | 25.68     | 4.69      |
|                 |                | 4             | 30       | 72.37       | 24.88     | 4.54      |
| <b>Separate</b> | <b>Harm</b>    | 0             | 60       | 97.47       | 5.48      | 0.71      |
|                 |                | 1             | 60       | 75.63       | 24.10     | 3.11      |
|                 |                | 2             | 60       | 71.43       | 30.60     | 3.95      |
|                 |                | 3             | 60       | 60.70       | 29.13     | 3.76      |
|                 |                | 4             | 60       | 52.47       | 30.51     | 3.94      |
|                 | <b>Help</b>    | 0             | 60       | 89.85       | 16.91     | 2.18      |
|                 |                | 1             | 60       | 74.75       | 26.43     | 3.41      |
|                 |                | 2             | 60       | 80.57       | 22.08     | 2.85      |
|                 |                | 3             | 60       | 68.70       | 25.58     | 3.30      |
|                 |                | 4             | 60       | 74.32       | 22.67     | 2.93      |